

DAT NGUYEN CONG

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RESEARCH INTERESTS

My research focuses on developing generative models that are robust to noise, training-inference alignment and computational efficient. I have a primarily interests in diffusion models, particularly **Diffusion Language Model**, **Multimodal Learning**, and **Time Series Analysis**, and I am eager to explore related research directions.

EDUCATION

B.A. in Data Science and Artificial Intelligence,
Hanoi University of Science and Technology
Cumulative GPA: 3.8/4.0

Oct 2020 - Oct 2024

RESEARCH AND WORK EXPERIENCE

AI Research Resident

FPT Software AI Center

Aug 2023 - Aug 2026

Hanoi, Vietnam

- Develop efficient inference methods for diffusion-based language generation, reducing generation steps while preserving strong sequence-to-sequence performance.
- Study test-time reasoning dynamics in autoregressive models, including architecture behavior and “overthinking”, and develop early-stopping mechanisms to terminate unhelpful computation.
- Investigate how noisy or contaminated data affects diffusion models and develop lightweight post-training guidance methods that improve generation quality with limited inference overhead.

AI Engineer Intern

FPT Camera, FPT Telecom

Jun 2023 - Feb 2024

Hanoi, Vietnam

- Developed a real-time human tracking and counting pipeline combining object detection and multi-object tracking; tested it on Viettel and Hikvision camera streams over wireless and local deployments.
- Built an Android prototype for pedestrian counting and reporting, and analyzed tracking failures caused by small or distant targets and identity switches.

Research Assistant

Data Science Laboratory

Advisors: Khoat Than and Tung Doan

Dec 2021 - Aug 2024

BKAI, HUST

- Studied reconstruction-based medical anomaly detection and diffusion guidance for preserving normal structures while repairing anomalous regions; the proposed approach outperformed the evaluated baselines.
- Investigated efficient training for distance-based sequence alignment with optimal transport and identified instability in expectation-maximization-based hyperparameter optimization.

PUBLICATIONS

(*) denotes equal contributions.

1. Tuan-Binh Tran*, **Dat Nguyen-Cong***, Duc-Trong Le, Thanh Trung Huynh, Tung Kieu. [ScenarioDiff: A Scenario-level Guidance Framework for Multimodal Time Series Forecasting](#). *International Conference on Data Mining (ICDM 2026)*.
2. **Dat Nguyen-Cong**, Tung Kieu, Hoang Thanh-Tung. [FastDiSS: Few-step Match Many-step Diffusion Language Model on Sequence-to-Sequence Generation](#). *Annual Meeting of the Association for Computational Linguistics (ACL Findings 2026)*.
3. **Dat Nguyen-Cong**, Hieu Tran Bao, Hoang Thanh-Tung. [Guiding Noisy Label Diffusion Models with Score-based Discriminator Correction](#). *International Conference on Computer Vision (ICCV 2025)*.

4. Luong Tran, Lan-Cuong Nguyen, Huynh Dang Nguyen, **Dat Nguyen-Cong**, Dung D. Le, Van Nguyen. [CLIP-FMoE: Scalable CLIP via Fused Mixture-of-Experts with Enforced Specialization](#). *International Conference on Learning Representations (ICLR 2026)*.

PREPRINTS

(*) denotes equal contributions.

1. **Dat Nguyen-Cong**, Luong Tran, Tung Kieu. [When Denoising Hurts: Rethinking the Terminal Step of Diffusion Time Series Forecasters](#). *Under review*.

2. Thanh Duc Pham, Lan-Cuong Nguyen, **Dat Nguyen Cong**, Dung D. Le, Minh-Tan Pham. Balancing Fidelity and Diversity of Synthetic Data Representations in CLIP-based Few-Shot Learning. *Under review*.

3. Hieu Tran Bao*, **Nguyen Cong Dat***, Nguyen Duc Anh, Hoang Thanh-Tung. [Learning to Stop Overthinking at Test Time](#). *Preprint*.

4. **Dat Nguyen Cong**, Hoang Thanh-Tung. On the Effect of Noisy Data in Diffusion Models. *Submitted*.

AWARDS AND ACHIEVEMENTS

Best Presentation Award <i>School of Information and Communication Technology, HUST</i> Awarded for the students with outstanding thesis presentation.	2024
Third Prize, Vietnamese National Olympiad in Physics <i>Tran Phu High School for the Gifted</i> The most prestigious physics competition for high school students in Vietnam.	2019
Gold Medal, Regional Contest in Coastal and Northern Delta Areas – Physics <i>Tran Phu High School for the Gifted</i> Physics competition among specialized high schools in the Coastal and Northern Delta regions.	2019

PRESENTATIONS

1. FastDiSS: Few-step Match Many-step Diffusion Language Model on Sequence-to-Sequence Generation. Poster Presentation. *ACL 2026*.

2. Guiding Noisy Label Diffusion Models with Score Discriminator Correction. Poster Presentation. *ICCV 2025*.

LANGUAGES

- **Vietnamese:** Native
- **English:** Professional working proficiency, IELTS 7.0

PROFESSIONAL SERVICES

Reviewer: ICML 2026, NeurIPS 2026.

SKILLS

Programming	Python, Java, Codex
Technical	Math, Statistics, Machine Learning
Libraries	PyTorch, Huggingface, OpenAI